HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented backend services for an AI-assisted authoring experience, translating natural language prompts into machine-readable plans for agentic orchestration.
- Integrated **large language models** into the planning service to generate structured outputs, handling non-deterministic behaviour and enforcing schemas for plan validation, reducing invalid plan generation by **40%**.
- Developed RESTful APIs and microservices using **Java Spring Boot**, focusing on object-oriented principles and design patterns to support core platform functionality.
- Built data pipelines using **Python** and **pandas** to integrate and normalise data from disparate internal services, creating a common model for use in plan assembly and discovery.
- Managed CI/CD pipelines using **Jenkins** and **Git** for automated testing and deployment of backend services, contributing to a **20%** reduction in deployment lead time.
- Collaborated within an agile team, actively participating in code reviews and sprint planning to align on technical direction and estimate feature delivery for agentic orchestration.
- Containerised backend services using **Docker** and deployed them to an **AWS** environment, ensuring consistent runtime configurations and facilitating scalability.

SKILLS

Backend Development – Java, Python, Spring Boot, REST APIs, Microservices, OOP, Design Patterns, Data Structures, Algorithms, Flask, FastAPI

AI/LLM Integration – Large Language Models, Structured Output Generation, Non-deterministic Systems, Schema Enforcement, Plan Generation

Data Engineering – Python, pandas, NumPy, ETL, Data Pipelines, Data Integration, Data Normalisation

Cloud & DevOps – AWS, Docker, Kubernetes, Jenkins, Git, CI/CD

Methodologies – Agile, Scrum, Code Reviews, Stakeholder Alignment

PROJECTS

AI Plan Generation Service (HCL Technologies)

- Developed a Python-based service to ingest user intents and generate structured execution plans for an automated workflow system, processing approximately **500 requests per hour**.
- Integrated with a **large language model API** to translate high-level natural language goals into detailed, machine-readable steps, implementing retry mechanisms for malformed responses.
- Designed and implemented RESTful endpoints using **FastAPI** for plan retrieval and validation, ensuring smooth integration with downstream orchestration services.

Enterprise Data Integration Hub (HCL Technologies)

- Built Python scripts to extract, transform, and load data from multiple enterprise systems (PostgreSQL, Oracle) into a central data repository, supporting operational reporting.
- Developed data normalization logic to reconcile disparate data formats and schemas, improving data consistency for downstream analytics by an estimated **30%**.
- Managed data pipeline execution using scheduled jobs and basic monitoring, ensuring timely data availability for business analysts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)